

Cathode-ray Tube :

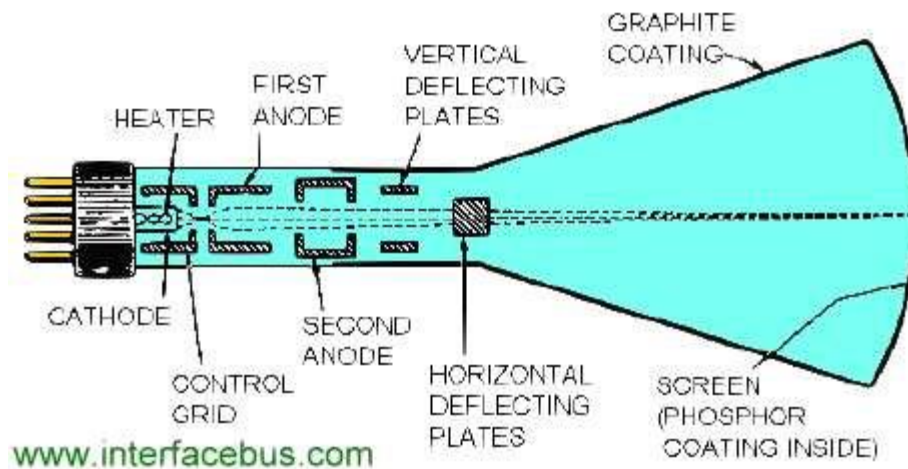


Fig : Basic Construction of Cathode Ray Tube with electrostatic deflection

CRT: Cathode-ray Tube is used to display small dot pattern on the screen. The Small dot on the screen is called as **pixel** or **pel**. Pixel is the smallest addressable point on the screen.

A Cathode-ray Tube is used as computer monitors, television screens or radar displays. A CRT is assembled into a monitor.

CRT consists of

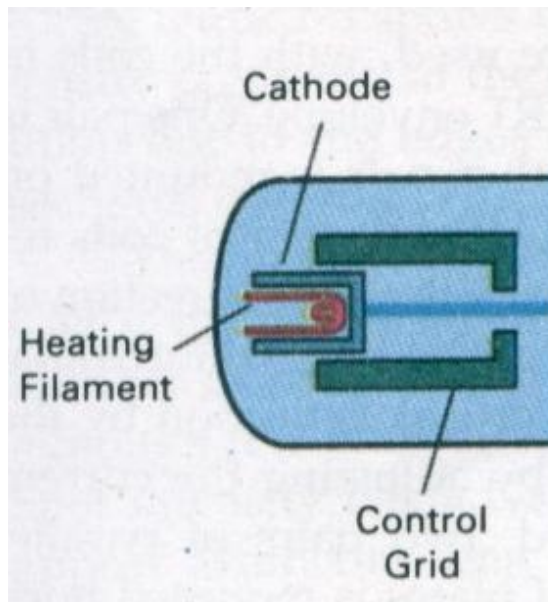
1. Electron Gun
2. Focussing System
3. Accelerating System
4. Deflection System
5. Screen with Phosphor coating

1. **Electron Gun** : It generates a beam of electrons which passes through focussing & defection system and strikes on specified positions on phosphor coated screen.

When electron beam strikes phosphor, that phosphor spot emits light and thus the point is visible.

It has_

- a) **Heating Filament** : Usually a wire or ribbon, which is heated by current passing through it.
- b) **Cathode** : The electrode that is the source of electron flow.
- c) **Control Grid** : Cathode is surrounded by metal control grid with hole at one end which allows electrons to escape.



Intensity of electron beam is controlled by setting voltage level on control grid.

A high $-ve$ voltage applied to control grid will shut off beam by repelling and stopping them from passing through small hole of control grid.

2. **Focusing System** : A pair of electrode of a CRT that is used to focus the electrons into a tight beam. It converges the small spot of electrons as it strikes on the screen. Otherwise beam will spread when it reaches the screen because electron repels each other.

Focusing is done by either magnetic or electric field.

3. **Accelerating System** : It consists of two metal plates mounted perpendicular to beam axis to accelerate beam to necessary velocity.
4. **Deflection System** : Two parallel plates/coils used to bend an electron beam a desired amount.

Horizontal-Deflection Plates. A pair of parallel electrodes that moves the electron beam from side to side in a CRT.

Vertical Deflection Plates. A pair of parallel electrodes in a CRT that moves the electron beam up and down.

For electromagnetic deflection two pairs of magnetic coils are mounted outside the tube neck.

5. **Screen** : The surface of a cathode-ray tube that produces a visible image or pattern. When electrons in the beam collide with phosphor coating they are then stopped. Screen absorbs the kinetic energy of the electrons.

Advantages:

- The cathode ray tube can easily increase the monitor's brightness by reflecting the light.
- They produce more colours
- The Cathode Ray Tube monitors have lower price rate than the LCD display or Plasma display.
- The quality of the image displayed on a Cathode Ray Tube is superior to the LCD and Plasma monitors.
- The contrast features of the cathode ray tube monitor are considered highly excellent.

Disadvantages:

- Produces x-rays; some are not blocked by lead shield
- Disposal: have to deal with lead; illegal to just dump in trash
- Power hungry (110 watts; compared to 35 watt LCDs). Accounts for 80% of electricity use in typical home computer setup!
- CRTs operate at very high voltage which can overheat system.
- They are Big, bulky & heavy to pick up and carry around

Random-scan technique :

Random scan monitors draw a picture one line at a time and for this reason are also referred to as **vector displays** (or stroke-writing or calligraphic displays).

The component lines of a picture can be drawn (Figure 2) and refreshed by a random - scan system in any specified order. Refresh rate on a random scan system depends on the number of lines to be displayed.

Picture definition is now stored as a set of line-drawing commands in an area of memory referred to as the **refresh display file**. To display a specified picture, the system cycles through the set of commands in the display file, drawing each component line in turn.

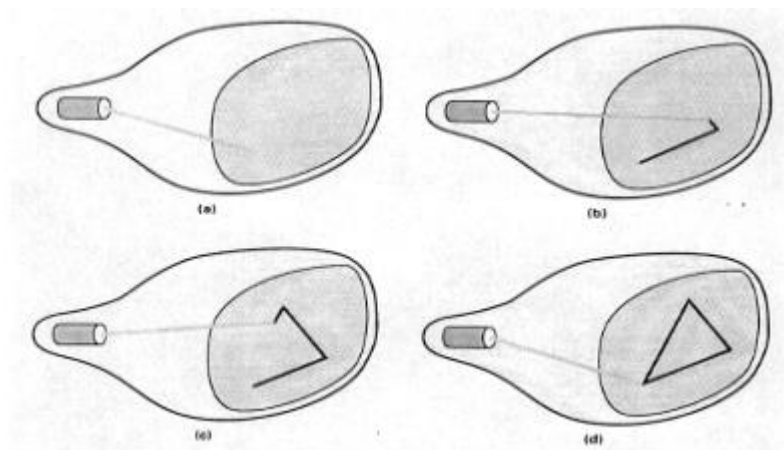
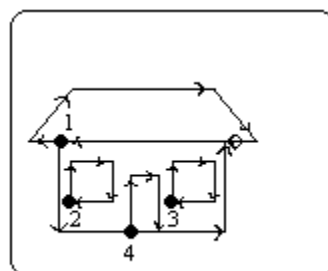


Fig 2 : Random-scan system

- Beam path directly controlled by vector commands; e.g.,
 - *moveto*(x, y)
 - *draw*(x, y)

Vector Based



Advantages and Disadvantages of raster scan system.

- **Advantages**
 - Low cost (memory has become cheap)
 - Refresh rate independent of image complexity
 - Can handle colour and filled areas
- **Disadvantages**
 - Uses more memory
 - Mathematical models of objects must be scan converted by the host processor
 - aliasing
 - jagged lines that are plotted as discrete points

Advantages and disadvantages of random scan system.

Advantages

- It gives high resolution since picture definition is stored as line drawing commands, easy animation, and requires little memory (just display program),

Disadvantages

- It gives limited colour capability and flicker occurs as complexity of image increases

Vector scan

► Strength

- Well applies to wfm monitors and vector scopes.
- Real time application
- Analog memory by Phosphor persistence
- fast response (350 ps step response in 1GHz scope)

► Weaknesses

- With time the cost of memory went down, the manufacturing cost of electrostatic display increased substantially.
- Required colors on screen.
- More information required, for a more complex display has made vector scan circuits more expensive and complex.

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Raster-scan technique :

In a raster-scan system, the electron beam is swept across the screen, one row at a time from top to bottom. As the electron beam moves across each row, the beam intensity is turned on and off to create a pattern of illuminated spots.

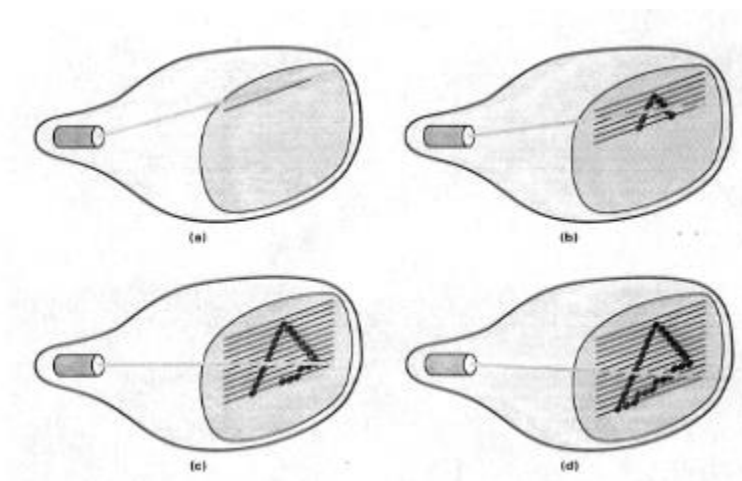
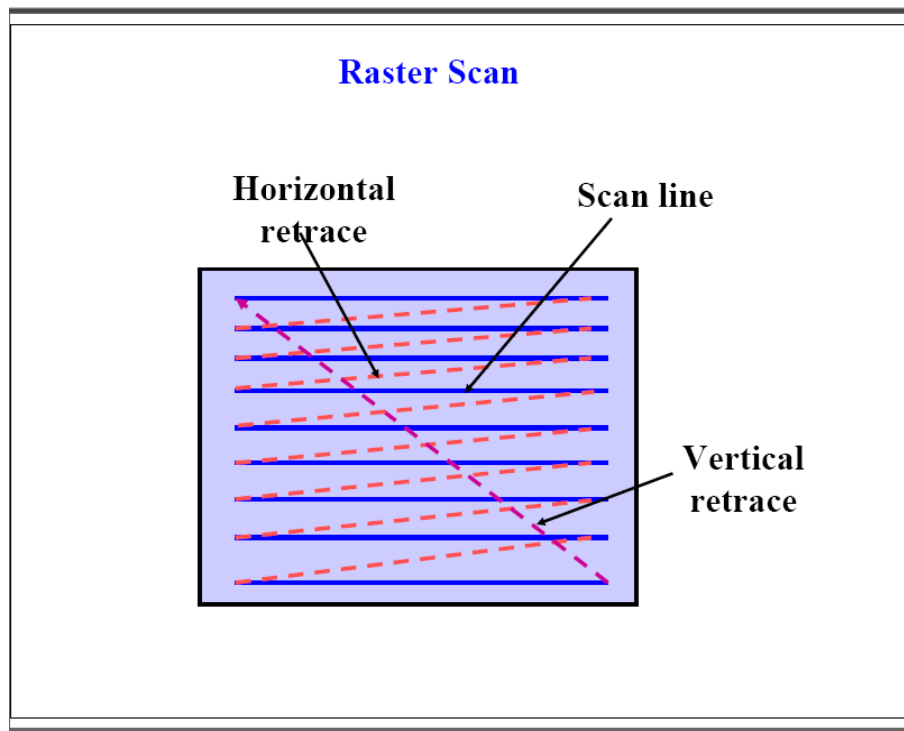


Fig 1 : Raster-scan system

Picture definition is stored in memory area called the **refresh buffer or frame buffer**. This memory area holds the set of intensity values for all the screen points. Stored intensity values are then retrieved from the refresh buffer and “painted ” on the screen one row (scan line) at a time (Fig. 1).

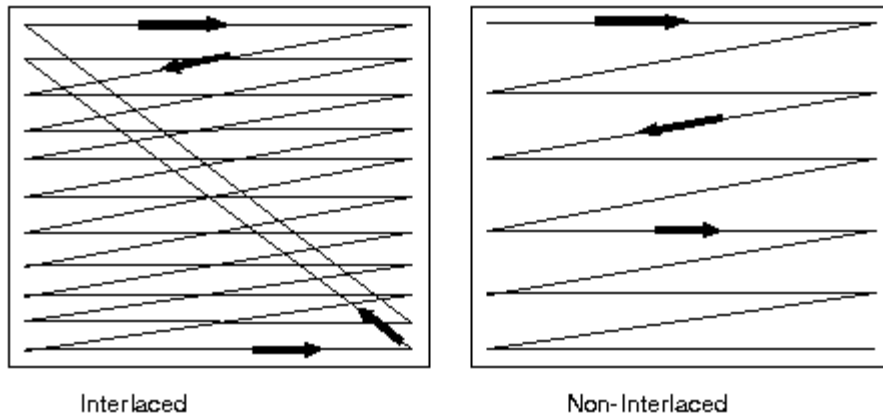
Refreshing on raster -scan displays is carried out at the rate of **60 to 80 frames per second**, although some systems are designed for higher refresh rates. Sometimes, refresh rates are described in units of cycles per second, or Hertz (Hz), where a cycle corresponds to one frame.



At the end of each scan line, the electron beam returns to the left side of the screen to begin displaying the next scan line. The return to the left of the screen, after refreshing each scan line, is called the **horizontal retrace** of the electron beam.

And at the end of each frame (displayed in 1/80th to 1/60th of a second), the electron beam returns to the top left corner of the screen to begin the next frame is called the **vertical retrace**.

On some raster -scan systems (and in TV sets), each frame is displayed in two passes using an **interlaced refresh procedure**. In the first pass, the beam sweeps across every other scan line from top to bottom. Then after the vertical retrace, the beam sweeps out the remaining scan lines (fig.below).



Interlacing of the scan lines in this way allows us to see the entire screen displayed in one-half the time it would have taken to sweep across all the lines at once from top to bottom.

Interlaced



Odd lines
Field 1



Even lines
Field 2



Field 1 + Field 2 = Frame (complete image)

Display Rate: 60 fields per second (North America)

Raster Scan

► Strength

- To each pixel is possible to add chrominance information
- The frame and horizontal SCAN is set by the screen resolution (1024x728). We talk about pixels and no more about vectors.
- the information resides in the luminance intensity of each pixel.
- Position on screen is predefined by the scan process.
- Less expensive than vector display
- The DSP and graphic processors are cheap and extremely powerful
- Very efficient to represent full images

► Weaknesses

- Slow screen update rate, normally 25-120 screen/sec
- Extensive DSP required to achieve analog persistence simulation
- At low resolutions pixels are quite "big"
- NON real time display
- Improper Interpolation of digital samples can produce visual artifacts (aliasing)

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Differentiate between Random and Raster Scan Display:

Random Scan	Raster Scan
1. It has high Resolution	1. Its resolution is low.
2. It is more expensive	2. It is less expensive
3. Any modification if needed is easy	3.Modification is tough
4. Solid pattern is tough to fill	4.Solid pattern is easy to fill
5. Refresh rate depends or resolution	5. Refresh rate does not depend on the picture.
6. Only screen with view on an area is displayed.	6. Whole screen is scanned.
7. Beam Penetration technology come under it.	7. Shadow mark technology came under this.
8. It does not use interlacing method.	8. It uses interlacing
9. It is restricted to line drawing applications	9. It is suitable for realistic display.

Shadow Mask Colour CRT :

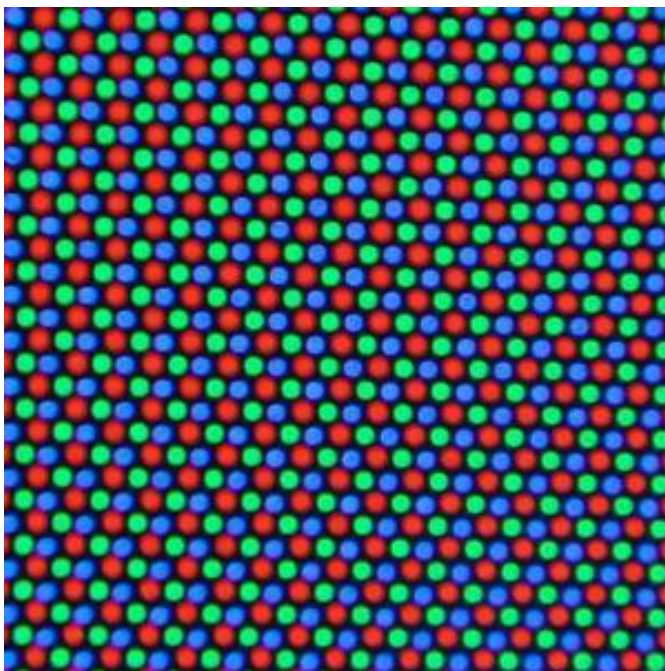
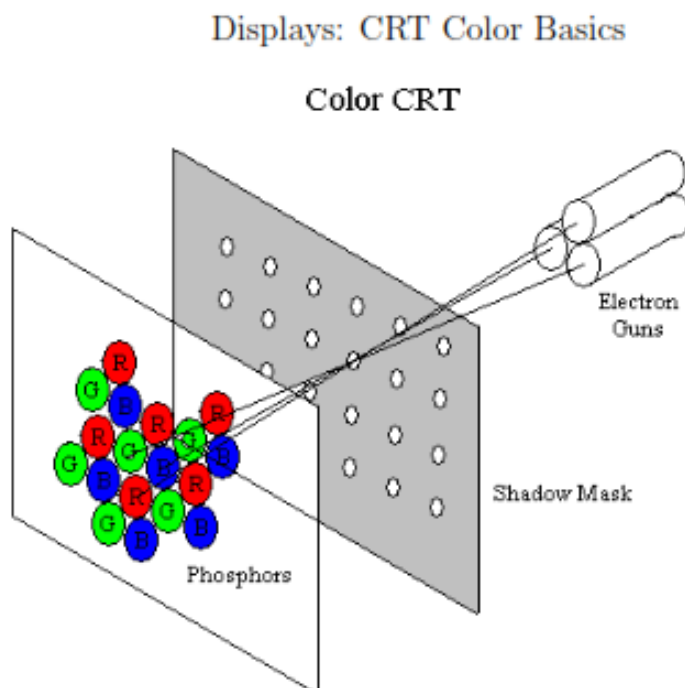
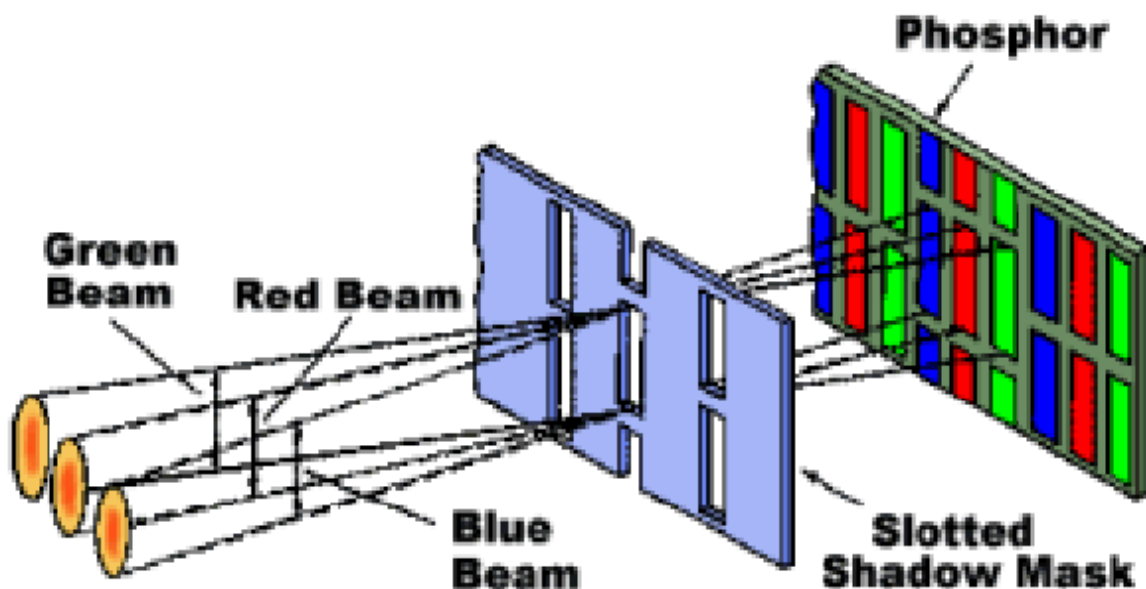


Fig : Screen with Red, Green & Blue colour phosphor forming a pixel.

- Color CRT uses 3 phosphors: RGB
- Uses 3 electron guns, one for each color phosphor
- One pixel consists of 3 points of phosphor
 - Basic arrangement is triangular, with one color at each apex
- Must be able to illuminate each pixel individually
 - Problematic due to characteristics of electron gun and accuracy
- Shadow mask
 - Very thin perforated sheet behind screen
 - Has one hole per pixel
 - Each hole is guide for beam to a single pixel
 - * Blocks any part of beam that would strike elsewhere
- Basic configuration is delta-delta
 - Triad of phosphors, triad of guns

The in-line method:

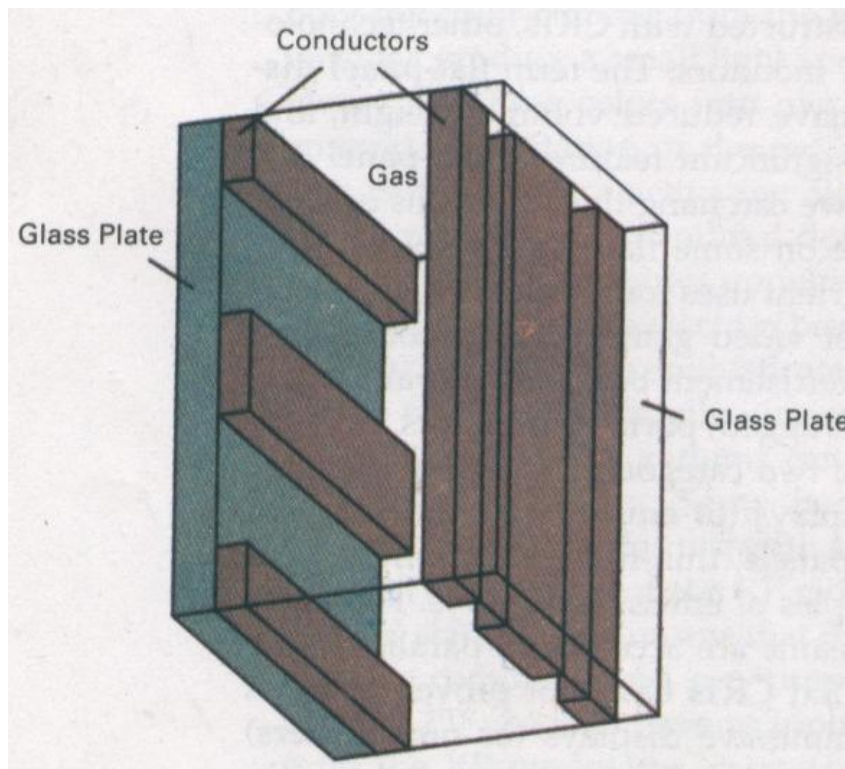


Flat-Panel Displays

- A class of video devices that have reduced volume, weight and power requirement compared with CRT
- Two main categories
 - Emissive Displays
 - Convert electrical energy to light energy
 - e.g. Plasma panels
 - Non-emissive Displays
 - Use optical effects to convert light from other sources into graphics patterns
 - e.g. LCD monitors

Plasma panels (gas-discharge display)

- Constructed by filling the region between two glass plates with a mixture of gases
- Refresh buffer used to store picture information
- Firing voltages applied to refresh the pixel positions



It consists of two sheets of glass with thin and closed spaced gold electrodes attached to inner faces and covered with dielectric material.

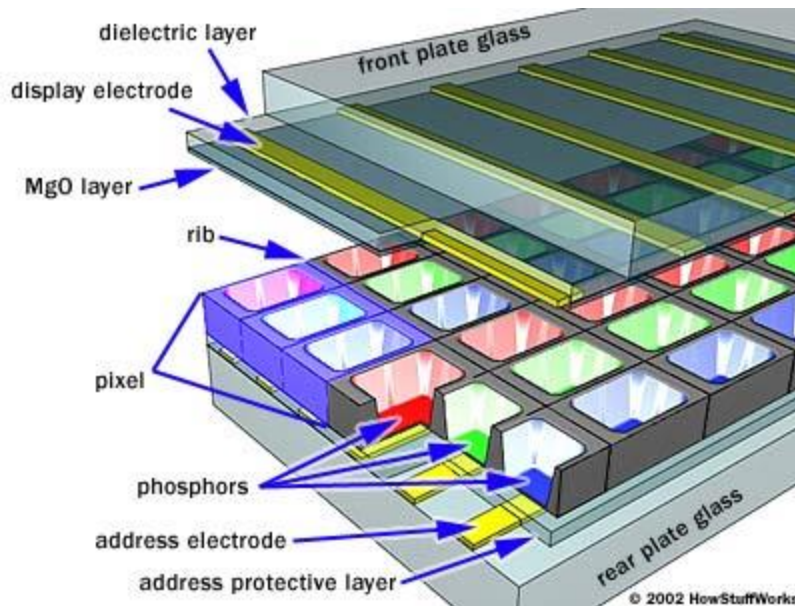
These are attached as a series of vertical conductors on one glass plate and horizontal on other side glass plate.

A space between these two glass plates is filled with a neon/xenon based gas mixture and sealed.

The plates are sealed so that the electrodes form right angles, each intersection creating pixels. When a voltage pulse passes between two electrodes, a small section of the gas, at the intersection of the conductors, break down into glowing plasma of electrons and ions, which emits UV radiation.

Phosphors are substances that give off light when they are exposed to other light.

The UV radiation activates color phosphors and visible light is emitted from each pixel.



Every **pixel** is made up of three separate **subpixel** cells, each with different colored phosphors. One subpixel has a red light phosphor, one subpixel has a green light phosphor and one subpixel has a blue light phosphor. These colors blend together to create the overall color of the pixel.

By varying the pulses of current flowing through the different cells, the control system can increase or decrease the intensity of each subpixel color to create hundreds of different combinations of red, green and blue. In this way, the control system can produce colors across the entire spectrum.

<http://electronics.howstuffworks.com/plasma-display2.htm>

Disadvantages of plasma

- Compared to other display units the plasma display unit has a very short life span, the manufactures estimate the life span to be around 20,000 hours to 30,000 hours (at a rate of 4 hours of TV a day that gives only 13.7 years).
- As your plasma display unit gets older its brightness gets dimmer.
- Plasma display units are considerably more expensive than cathode ray tube monitors.
- Plasma display units must be handled carefully because they are a very fragile display unit.

Advantages of plasma

- Larger viewing angle, 160° compared to LCD 40° & rear projection 120°. (Allows a larger audience to be able to view the image reproduction)
- No projection throw distance limitations.
- All plasma display units are considerably thinner in width than the cathode ray tube monitors.
- Most plasma display units are either free standing or can be mounted on a ceiling or wall.
- The plasma display units has a clearer image, brighter viewing angle, better colour quality and higher contrast ratio than the CRT and LCD display units.

Liquid-crystal Displays

Used in small systems, such as calculators, laptop computers.

This display uses nematic (thread like) liquid crystal compounds that tends to keep the long axes of rod shaped molecules aligned.

These nematic compounds have crystalline arrangement of molecules, yet they flow like a liquid and hence termed as liquid crystal display.

Liquid crystal material is filled in between two glass plates.

One glass plate has horizontal transparent conductors built into it and other end glass plate has vertical transparent conductors built into it.

The intersection of these two conductors defines the pixel position.

Polarized light passing through material is twisted so that it will pass through opposite polarizer.

The light is then reflected back to the viewer.

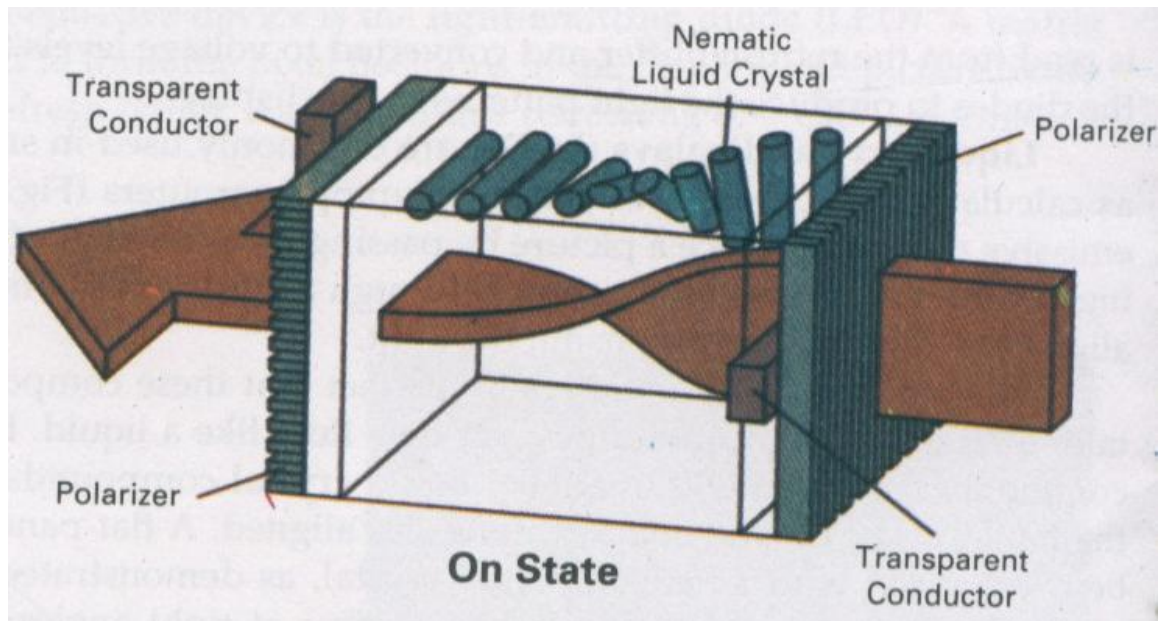
To, turn off pixel, we apply voltage to two intersecting conductors to align molecules so that light is not twisted.

The picture definition is stored in the refresh buffer and screen is refreshed at the rate of 60 frames/sec.

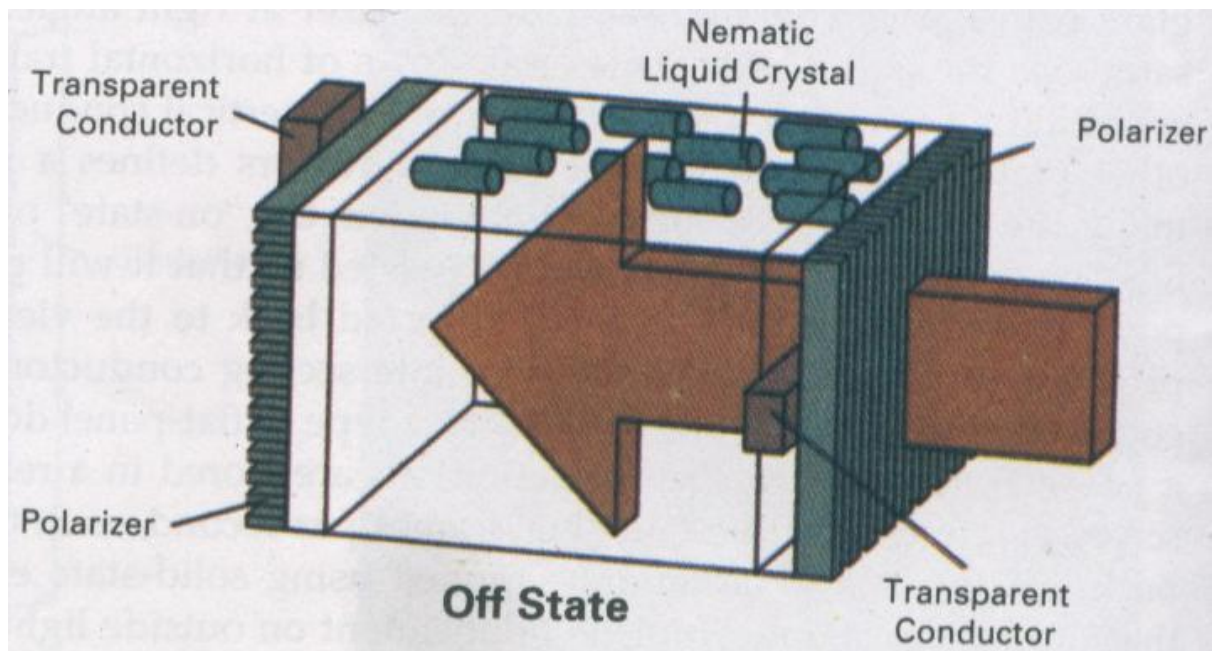
Produce a picture by passing polarized light (from the surrounding or from an internal light source) through a liquid-crystal material that can be aligned to either block or transmit the light.

Liquid crystal: These compounds have a crystalline arrangement of molecules, yet they flow like a liquid.

- Two glass plates, each containing a light polarizer at right angles to the other plate, sandwich the liquid crystal materials.
- Rows of horizontal transparent conductor & columns of vertical conductors (put into glass plates)

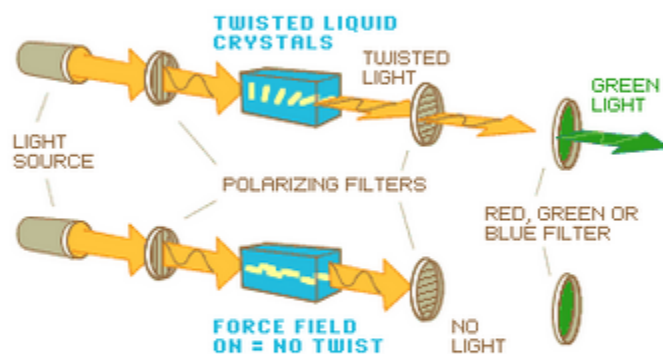


- Polarized light passing through the material is *twisted* so that it will pass through the opposite polarizer.
- The light is then reflected back to the viewer.



- To turn off the pixel, we apply a voltage to the two intersecting conductor to align the molecules so that the light is not twisted.

In the so-called twisted nematic (or TN) display, the molecules are arranged in this way. More specifically, the glass surfaces are treated such that the molecular direction is parallel to the admitting direction of each neighboring polarizer. Because these directions are crossed, the molecular direction is confined to a 90° twist from one side of the cell to the other (see figure).



In this case, what happens is that the light vibration follows this twist from one polarizer to the other, so that all light in fact passes the cell, without being absorbed, in spite of the fact that the polarizers are crossed. Hence, the cell appears bright.

As mentioned earlier, liquid

crystals are sensitive to electric forces. If you apply an electric field that is strong enough across a liquid crystal of the right kind, its molecules arrange themselves parallel to the electric field. So now, by applying a voltage across the liquid crystal cell, i.e. along the light direction, you destroy the twist and instead force the molecules into the direction in which they do not affect the polarization state of the light. All light is now absorbed by the crossed polarizers and the cell appears black when the electric field is turned on.

By creating a matrix of squares that locally control the state of the twist in their respective area you get a liquid crystal display containing a large number of individual picture elements (pixels).

Advantages of LCD

- The sharpness of a LCD display is at maximum tweakness.
- Zero geometric distortion at the native resolution of the panel.
- High peak intensity produces very bright images. Best for brightly lit environments.
- Screens are perfectly flat.
- Thin, with a small footprint. Consume little electricity and produce little heat
- The LCD display unit is very light and can be put anywhere or moved anywhere in the house.

Lack of flicker and low glare reduce eyestrain.

Disadvantages of LCD

- After a while the LCD display the some of the pixels will die you will see a discoloured spot on a black spot on the display.
- The cost of a LCD is considerably at a high price.
- The LCD display will have slow response times.
- The LCD display has a fixed resolution display and cannot be changed.
- LCDs use analog interface making careful adjustment of pixel tracking/phase in order to reduce or eliminate digital noise in the image.
- The viewing angle of a LCD display is very limited due to the Automatic pixel tracking/phase controls.

Working Principle of LED Display

LED DISPLAY UNIT:

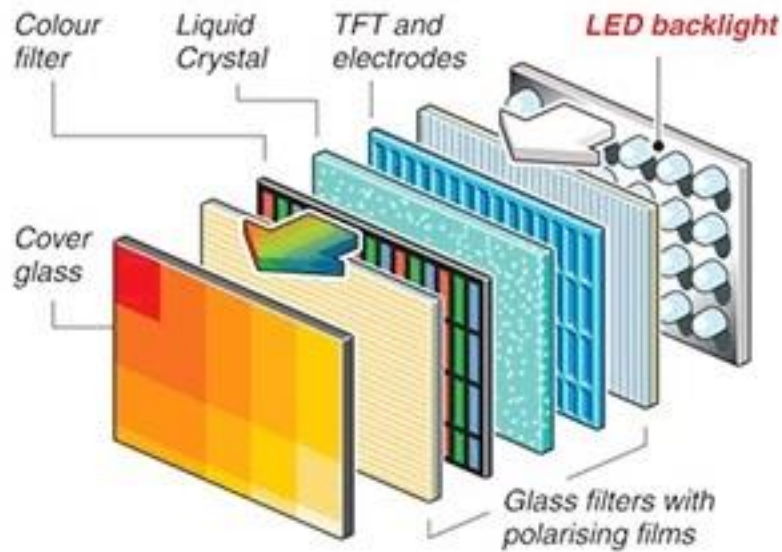
While both types of TVs use LCD technology and feature flat-panel designs, LED TV is a more sophisticated form of LCD TV. LED TVs use LED backlighting to illuminate LCDs, making them smaller, brighter, and more detailed than standard LCD TVs. However, not all LED televisions have the same backlighting.

LED television sets have redefined the watching experience in a whole new and improved way. LED displays are even smaller and lighter than LCDs. The operating mechanism of LED televisions is similar to that of LCD televisions. LED televisions may be thought of as a more advanced version of LCD televisions. LED TVs have reduced the disadvantages of LCD TVs.

The fluorescent bulbs in the LCD television took up a lot of space and were heavy, and they didn't always give good colour quality. These flaws have been overcome by LED televisions, which employ an array of LEDs to produce backlighting, as depicted in the figure. LED televisions take up less space than LCD televisions and are thus smaller and lighter. The LEDs may also display the most accurate black-and-white picture.

There are two types of LED TVs on the market right now. One is backlit, and it gets its backlighting from a slew of LEDs mounted on the back of the television display. The other is edge lighted, which gets its illumination from LEDs positioned around the borders of the television display. Edge-lit LED TVs are significantly slimmer than back-lit LED TVs. As a result, edge-lit LED TVs use less energy and are more efficient.

LED TVs use light-emitting diodes to illuminate their LCD panels. LEDs are made up of tiny semiconductors that light when exposed to electricity. The LED TV may change the alignment of the liquid crystal when an electric current is provided to a tiny specified location of the liquid crystal layer.



General concept of LED

This current travels between LED anodes and LED cathodes, which are positively and negatively charged electrodes, respectively. A conventional LCD TV, on the other hand, is backlit by fluorescent lights. These lamps work by emitting ultraviolet rays from mercury vapour, which light up the lamp's phosphor coating.

Unlike pure LED systems, LED-backlit LCDs are self-illuminating. There are several ways to backlight an LCD panel with LEDs, including using white or RGB (Red, Green, and Blue) LED arrays behind the panel, and edge LED lighting, which uses white LEDs arranged around the inside frame of the TV and a light diffusion panel to spread the light evenly behind the LCD panel.